Priva-Sync Task

Who doesn’t use generative AI these days?

This project too revolves around generation—specifically, predicting the next word using a federated machine learning system. However, before diving into that,it's important for us to know how basic applications are built and believe us , this task is going to give you a good hold on that!

We know you are still trying to learn through the resources provided , hence we give you ample time till **14th April EOD** to attempt this task.

We value the **learning process** and genuine effort put in much more than the mere completion of the task by using GenAI. To track and evaluate the same, we’ll be reviewing your **Git commit history**, which should reflect the step-by-step development of your work. Therefore, applicants are required to use Git commits to log each stage of development of the project. For example, let's say you just finished making the APIs for authentication, then your commit should be something along the same lines as : ‘add: authentication routes‘.

It is mandatory to have a README.md which is descriptive and explains your submission.

Vibe coded submissions are strictly discouraged.

Task : Build a **chatbot application** using any free API (e.g., Gemini, OpenAI, etc.) that can only be accessed by **authorized users**.

Try to do some research about which API will best suit your cause , which database will give you the easiest operations as you can be questioned upon the same(including your choice of frameworks) after your submissions.

Frontend :
Make an interface where users can
1.Signup
2.Login
3.Enter prompts and receive responses from the chatbot in a **chat-style interface.** Ensure proper handling and formatting of very large prompts and responses.The most basic requirement is handling text-based prompts and responses. If you are able to incorporate images, videos, or other forms of media, it will be considered a plus

Backend :
1.Create APIs to store user information (name, email, etc.) in a database
2.Implement **JWT-based authentication**
3.Properly configure **CORS**

Integration :
Finally , you are required to integrate your frontend and backend to make a fully functioning application.

Bonus :
1.Implement **session-based chat management**, allowing users to:

- View past chat sessions
- Reopen previous sessions and view conversation history

2.You are required to implement containerization for your application using a tool such as Docker.

Link for Resources : Resources

Finally, remember you don't need to feel overwhelmed by the resources. In case of any doubts , reach out to us :

- Ritika Batra : ritikab24@iitk.ac.in | 9528477602
- Yatin Bhojwani : yatinb24@iitk.ac.in | 9322741860
- Shivansh Jaiswal: shivanshj24@iitk.ac.in | 9971104638
- Harsh Malgatte: mharsh24@iitk.ac.in | 7385840159
- Austin Shijo: austins24@iitk.ac.in | 8946061070
- Muragesh Nyamagoud: muragesh24@iitk.ac.in | 9353466456